8. In principle, there are two concepts of mass that enter into physics. One is the concept described in the text: mass as that property of an object which resists acceleration. Sometimes, this notion of mass is called *inertial mass.* The second concept of mass is the one relevant for gravity: mass as that property of an object which determines how strongly it will be pulled by a gravitational field of a specified strength (such as the earth's). Sometimes this notion of mass is called *gravitational mass.* At first glance, the Higgs field is relevant only for an understanding of inertial mass. However, the equivalence principle of general relativity asserts that the force felt from accelerated motion and from a gravitational field are indistinguishable—they are equivalent. And that implies an equivalence between the concepts of inertial mass and gravitational mass. Thus, the Higgs field is relevant for both kinds of mass we've mentioned since, according to Einstein, they are the same.

9. I thank Raphael Kasper for pointing out that this description is a variation on the prize-winning metaphor of Professor David Miller, submitted in response to British Science Minister William Waldegrave's challenge in 1993 to the British physics community to explain why taxpayer money should be spent on searching for the Higgs particle.

10. The mathematically inclined reader should note that the photons and W and Z bosons are described in the electroweak theory as lying in the adjoint representation of the group SU(2) × U(1), and hence are interchanged by the action of this group. Moreover, the equations of the electroweak theory possess complete symmetry under this group action and it is in this sense that we describe the force particles as being interrelated. More precisely, in the electroweak theory, the photon is a particular mixture of the gauge boson of the manifest U(1) symmetry and the U(1) subgroup of SU(2); it is thus tightly related to the weak gauge bosons. However, because of the symmetry group's product structure, the four bosons (there are actually two W bosons with opposite electric charges) do not fully mix under its action. In a sense, then, the weak and electromagnetic interactions are part of a single mathematical framework, but one that is not as fully unified as it might be. When one includes the strong interactions, the group is augmented by including an SU(3) factor—"color" SU(3)—and this group's having *three* independent factors, SU(3) × SU(2) × U(1), only highlights further the lack of complete unity. This is part of the motivation for grand